

Q. > Why Fourier Transform? / Limitations of Fourier Series

⇒ • ~~In time domain~~

- Only applicable for periodic function.
- Signal should be cont. w.r.t time but
or discrete w.r.t frequency.
- It is discrete function of freq.
- Function must follow Dirichlet condition.

∴ So we use Fourier transform.

Fourier transform

⇒ It can be written for any function provided -

- It is converging in nature $\left\{ \int_0^{\infty} f(t) dt = \text{finite. (not } \infty) \right\}$
- It is taken for any finite duration of time.

→ Applicable for both periodic & non-periodic function.

$$F.T[f(t)] =$$

$$F(j\omega) = \int_0^{\infty} f(t) e^{-j\omega t} dt$$

⇒ Fourier transform inverse.

$$F.T^{-1}(f(j\omega)) =$$

$$f(t) = \frac{1}{2\pi j} \int_{-\pi}^{\pi} F(j\omega) e^{j\omega t} d\omega$$

⊙ Properties -

- Distributive
- Linear.
- Commutative.

⇒ Convolution theorem

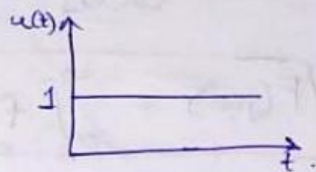
⇒ Fourier transform of linear convolution integral of 2 funct's is equal to the product of their respective Fourier transforms.

$$Y(e^{j\omega}) = X(e^{j\omega}) G(e^{j\omega})$$

$$\therefore Y(e^{j\omega}) = \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^t x(\tau) g(t-\tau) d\tau \right] e^{-j\omega t} dt$$

Consider "unit step function"

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$



$$\therefore Y(e^{j\omega}) u(t-\tau) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) g(t-\tau) u(t-\tau) d\tau \right] e^{-j\omega t} dt$$

$$\begin{aligned} (0 < \tau < t) & \begin{cases} \text{box } t > \tau \Rightarrow u(t-\tau) = 1 \\ \text{" } t < \tau \Rightarrow u(t-\tau) = 0 \end{cases} \end{aligned}$$

$$\begin{aligned} &= \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} g(t-\tau) u(t-\tau) e^{-j\omega t} dt \right] d\tau \\ &= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega \tau} \left[\int_{-\infty}^{\infty} g(t-\tau) u(t-\tau) e^{-j\omega(t-\tau)} dt \right] d\tau \end{aligned}$$

Let $t-\tau = m$, $dt = dm$

$$= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega \tau} \left[\int_{-\infty}^{\infty} g(m) e^{-j\omega m} dm \right] d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega \tau} G(e^{j\omega}) d\tau$$

$$= G(e^{j\omega}) \int_{-\infty}^{\infty} x(\tau) e^{-j\omega \tau} d\tau = G(e^{j\omega}) X(e^{j\omega})$$

$$\therefore \boxed{Y(e^{j\omega}) = X(e^{j\omega}) \cdot G(e^{j\omega})}$$

Continuous time Fourier transform (C.T.F.T)

evaluation prop. of Fourier transform.
Symmetric prop.

Conjugate Symm.

$$x^*(-t) = x(t)$$

Conjugate anti-symm

$$x^*(t) = -x(t)$$

$$R.P = \frac{1}{2} [x(t) + x^*(t)]$$

$$I.P = \frac{1}{2} [x(t) - x^*(t)]$$

$$C.Symm.P = \frac{1}{2} [x(t) + x^*(-t)]$$

$$C.ASymm.P = \frac{1}{2} [x(t) - x^*(-t)]$$

1. >

* It states that Fourier transform R.P of any complex function is = to conjugate Symm. part of that function.

* ~~Fourier transform of conjugate Symm part of any~~

~~comp. fun. is = to real symm part of Fourier transform of that function.~~

2. > F.T (imag.) = conjugate anti-symm. part of F.T.

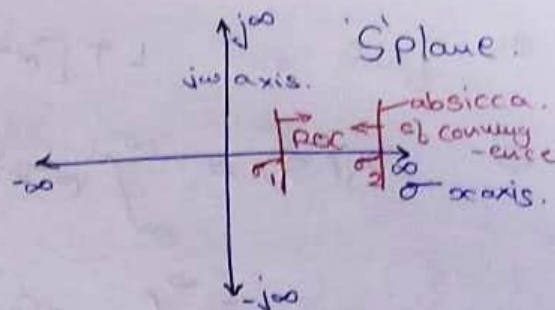
3. > F.T (conj. symm) = real part of F.T.

4. > F.T (conj. anti symm) = imag. part. of F.T

Laplace Transform :-

→ Used for which function - which are not converging in nature.

→ we use complex frequency $X(s)$.
where 's' is a complex frequency variable.



→ generalised way of frequency domain representation

- if $\sigma = 0$ then we get Fourier Series.
- if we add σ in Fourier transform $-\infty$ to ∞ then we get Laplace transform in a Region ROC.

$$X(s) = \int_0^{\infty} e^{-\sigma t} f(t) e^{-j\omega t} dt$$
$$= \int_0^{\infty} f(t) e^{-(\sigma + j\omega)t} dt$$

$$\therefore X(s) = \int_0^{\infty} f(t) e^{-st} dt$$

where $s \rightarrow$ complex frequency

⇒ Initially $f(t)$ is a non-converging function we multiply by decaying function $e^{-j\omega t}$, so that we convert it into converging function for some value of σ , then we can apply Fourier transform.

⇒ find out $LT[e^{-at}]$.

$$\Rightarrow X(s) = \int_0^{\infty} e^{-at} e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt = \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_0^{\infty}$$

$$= \frac{1}{s+a}$$

$$= \left[\frac{e^{-(s+a)t}}{s+a} \right]_0^{\infty}$$

~~inverse Laplace transform of $\frac{1}{s+a}$ is~~

if $a = -1$ $\therefore$ expand.

$$\therefore LT[e^t] = \frac{1}{s-1} \therefore = \frac{1}{s-1} = -1(1-s)^{-1}$$

$$= -1(1-s)^{-1}$$

$$\therefore \boxed{LT[t^n] = \frac{n!}{s^{n+1}}}$$

Some basic properties of Laplace transform

↳ time shifting :-

in time domain

= freq. funct. $\times$
expo. function.

$$f(t) \xrightarrow{LT} F(s)$$

$$f(t-t_0) \xrightarrow{LT} \int_0^{\infty} f(t-t_0) \cdot e^{-st} dt$$

$$= \int_0^{\infty} f(t-t_0) e^{-s(t-t_0)} \cdot e^{-st_0} dt$$

$$= e^{-st_0} \int_0^{\infty} f(t-t_0) e^{-s(t-t_0)} dt$$

$$= e^{-st_0} F(s)$$

2> Frequency Shifting ^(new) :-

3> Differentiation & Integration ^(new) :-

$$f(t) \xrightarrow{LT} F(s)$$

$$\frac{d f(t)}{dt} \xrightarrow{LT} s F(s) - f(0)$$

$$\therefore \int_0^{\infty} f(t) e^{-st} dt$$

$$= \left[f(t) \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} f(t) \frac{e^{-st}}{-s} dt$$

$$= f(0) \cdot \frac{1}{s} + \frac{1}{s} \int_0^{\infty} f'(t) e^{-st} dt$$

$$F(s) = \frac{f(0)}{s} + \frac{1}{s} LT [f'(t)]$$

$$\therefore s F(s) - f(0) = \frac{1}{s} LT [f'(t)]$$

Initial Value Theorem :-

$$\bullet \lim_{t \rightarrow 0} f(t) = f(0) = \lim_{s \rightarrow \infty} [s F(s)]$$

Prove

$$\Rightarrow \text{LT} \left[\frac{d}{dt} f(t) \right] = s F(s) - f(0)$$

$$\Rightarrow \int_0^{\infty} \frac{d}{dt} f(t) e^{-st} dt = s F(s) - f(0)$$

take limit $s \rightarrow \infty$.

$$\lim_{s \rightarrow \infty} \left[\int_0^{\infty} \frac{d}{dt} f(t) e^{-st} dt \right] = \lim_{s \rightarrow \infty} [s F(s) - f(0)]$$

$$0 = \lim_{s \rightarrow \infty} [s F(s)] - f(0)$$

$$\therefore f(0) = \lim_{s \rightarrow \infty} [s F(s)]$$

final value theorem :-

$$\bullet \lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} (s F(s))$$

Prove

$$\Rightarrow \text{LT} \left[\frac{d}{dt} f(t) \right] = s F(s) - f(0)$$

$$\therefore \Rightarrow \int_0^{\infty} \frac{d}{dt} f(t) e^{-st} dt = s F(s) - f(0)$$

take limits

$$\lim_{s \rightarrow 0} \int_0^{\infty} \frac{d}{dt} f(t) e^{-st} dt = \lim_{s \rightarrow 0} [s F(s) - f(0)]$$

$$\therefore \therefore [f(t)]_0^{\infty} = \lim_{s \rightarrow 0} [s F(s)] - f(0)$$

$$f(\infty) - f(0) = \lim_{s \rightarrow 0} [s F(s)] - f(0)$$

$$\therefore \therefore \underline{f(\infty) = \lim_{s \rightarrow 0} [s F(s)]}$$

$$\boxed{f'(t) = f(t)}$$

⇒ Convolution Theorem :-

⇒ Laplace transform of linear convolution integral of 2 functions is equal to the product of their respective Laplace transforms.

$$\text{if } \begin{aligned} x(t) &= X(s) \\ g(t) &= G(s). \end{aligned}$$

$$\therefore \boxed{LT[y(t)] = L.T. [x(t) * g(t)] = X(s) \cdot G(s).}$$

Prove

$$\therefore LT[y(t)] = Y(s) = X(s) G(s)$$

$$= \int_0^{\infty} y(t) e^{-st} dt$$

$$= \int_0^{\infty} \left[\int_0^t x(z) g(t-z) dz \right] e^{-st} dt$$

consider unit step function.

$$= \int_0^{\infty} \left[\int_0^{\infty} x(z) g(t-z) u(t-z) dz \right] e^{-st} dt$$

$$= \int_0^{\infty} x(z) \left[\int_0^{\infty} g(t-z) u(t-z) e^{-st} dt \right] dz$$

$$= \int_0^{\infty} x(z) e^{-sz} \left[\int_0^{\infty} g(t-z) u(t-z) e^{-s(t-z)} dt \right] dz$$

$$= \int_0^{\infty} \because t-z = m, dt = dm$$

$$= \int_0^{\infty} x(z) e^{-sz} \left[\int_0^{\infty} g(m) e^{-sm} dm \right] dz$$

$$= \int_0^{\infty} x(z) e^{-sz} G(s) dz = G(s) \int_0^{\infty} x(z) e^{-sz} dz$$

$$= G(s) \cdot X(s)$$

$$\therefore \boxed{LT[y(t)] = X(s) \cdot G(s)}$$

$$\Rightarrow \left[G(s) = \frac{Y(s)}{X(s)} \right]$$

- are defined as those values of complex freq. in plane at which transfer func'n tends to ∞ .
in s-plane

- $\therefore$ transfer function $= \frac{1}{s+a}$
 $\therefore \mathcal{L}^{-1}\left[\frac{1}{s+a}\right] = e^{-at}$
 $= e^{-\sigma t}$
 converging or
 decaying
 function
 means BIBO
 $\Rightarrow$ so we get more
 stability due
 to $-ve$ expo.

2nd transfer function = $\frac{1}{s-a}$; $LT^{-1}\left[\frac{1}{s-a}\right] = e^{at}$
 $= e^{+ve}$

if +ve real part comes.

then we get more unstable, unbounded region

• for a system to be stable.

Poles should have -ve real part.
 or it should lie left side of the plane.

3rd at $a=0$, origin. it is a unit step function
 • marginally stable

$= \frac{1}{s}$; $LT^{-1}\left[\frac{1}{s}\right] = e^0$
 $= 1$ now const.

but after integ.

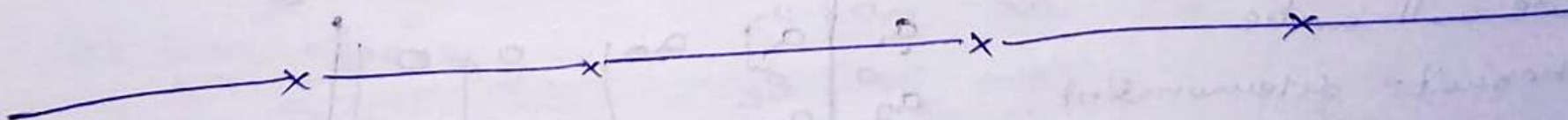
$\int_0^\infty 1 dt = \infty \rightarrow$ then unbounded/unstable

now. multiple order poles (it is always unstable).

$LT^{-1}\left[\frac{1}{s^2}\right] = e^t$

continuously $\uparrow$ too.
 unbounded region.

$t \Rightarrow 0 \rightarrow \infty$



- Criteria for finding a system is stable or not.

Let say

()

$$(s+a)(s+b)(s+c)(s^2+ds+e)$$

⇒ necessary condition for absolute stability of any system

⇒ All the powers of s must be present with non-zero.

co-eff. (from max. to min. powers.)

2> All these co-eff. must be +ve,

or All these co-eff. must be -ve,

we can

but if there is a change in sign → then indirectly state, i.e. unstable.

(upto 2nd order. These 2 above condition are sufficient also)

$$s^2+as+b, \quad s^2+as+b.$$

$$s = \frac{-a \pm \sqrt{a^2-4b}}{2}$$

now for cubic onwards.

$$\therefore s = -ve \text{ always.}$$

we apply.

Hurwitz Criteria

Let
denominator
polynomial

$$\Rightarrow a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0.$$

we will make

Hurwitz determinant

of order $n \times n$.

where n = highest degree.

$$\begin{array}{c|cccc} a_1 & a_1 & a_0 & 0 & 0 \\ a_3 & a_3 & a_2 & a_1 & a_0 \\ a_5 & 0 & a_4 & a_3 & a_2 \\ a_7 & 0 & 0 & 0 & a_4 \end{array}$$

(a_{2n-1})

all principal

minors of

determinant

should be +ve.

• then it will be absolute

Let say box take ex:- of 10th order. (means)

too so box simple evaluation.

∴ we need.

Routh Criteria :-

we need to calculate routh array
in which we need to compute 2 by 2 element.
determinant.

let take.

$$a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0$$

South array.

(max spaces)

s^4	a_0	a_2	a_4
s^3	a_1	a_3	0
s^2	$\frac{a_1 a_2 - a_3 a_0}{a_1}$	$\frac{a_1 a_4 - 0}{a_1}$	
s^1			
s^0			

rows of south array.

- how many times changes in sign is there in 1st column.
- ∴ we can say that, that no. of elements are on right hand side.
- for instability

let $\frac{a_1 a_2 - a_3 a_0}{a_1} = b_1$

$a_4 = b_2$

s^4	a_0	a_2	a_4
s^3	a_1	a_3	0
s^2	b_1	b_2	
s^1	$\frac{b_1 a_3 - b_2 a_1}{b_1}$		

let $\frac{b_1 a_3 - b_2 a_1}{b_1} = c_1$

s^4	a_0	a_2	a_4
s^3	a_1	a_3	0
s^2	b_1	b_2	
s^1	c_1		
s^0	d_1		

if all 1st column element's are +ve then my system is absolute stable system.

2 things ~~that~~ may happen which will make.

Routh's criteria bail -

(a) 1st condition

• when while comp. routh array.

if 1st element comes out be zero with some non-zero element in same row.

if $b_1 = 0$.
then it
will
make
 ∞ in next
column element

a_0	a_2	a_4
a_1	a_3	
b_1	b_2	
c_1		
d_1		

• To solve this condition, we ~~not~~ replace zero by +ve ϵ (epsilon)

then find apr. ϵ for that ~~not~~ same values (same elements)
calculate ~~for~~ ϵ in ϵ ki same values me.
 $\lim \epsilon \rightarrow 0^+$ Range, then in sub D_i value

a_0
b_1
ϵ
c_1
d_1

again same sign ϵ aati hai then it will marginally stable.
if not (means sign change) then it will absolute unstable.

(b) convert $f(s) \rightarrow f(z)$

by $z = \frac{1}{s}$. then co-efficient interchange.

then make new routh array.

then ~~then~~ there is no problem.

* now we get all +ve in 1st column.

then it will be marginally stable.

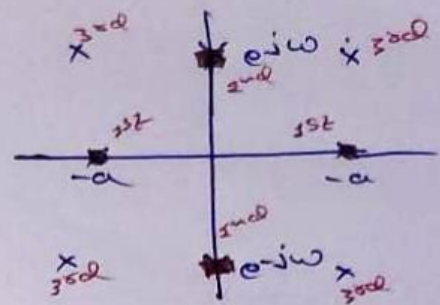
2nd Condition:

- if one complete row is zero.
then root may be only for a particular condition.

when (Symmetrical placed roots)

means mirror image.

It appears.



~~then~~ ~~the~~ row ~~is~~ ~~zero's~~ (this condition)

- ex:- S_3 is zero [only with $\pm$ (odd power) row] comes zero.
- then it will come S_4 (above S_3).

↑
Solution
make an
auxiliary
eq.

$$s^4 \quad a_1 \quad a_2 \quad a_3$$

$$s^3 \quad a_1 s^4 + a_2 s^2 + a_3 s^0$$

roots of this auxiliary eq.

if all roots comes out
in img. axis.

then it is
marginally stable.

if roots comes
on real axis
also, then it -
is absolute
unstable.

LAPLACE TRANSFORM

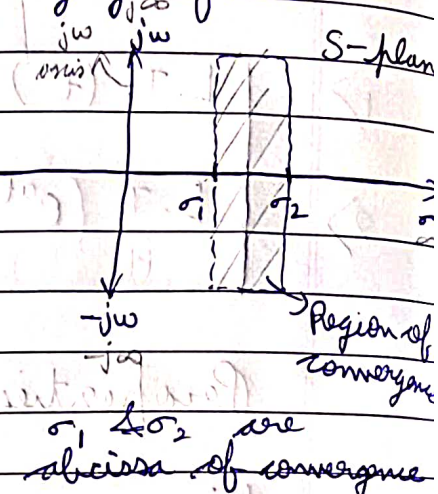
→ It can be used to write frequency domain function for a non-converging functⁿ

$$X(s) = \int_0^{\infty} e^{-\sigma t} f(t) e^{-j\omega t} dt$$

$$= \int_0^{\infty} f(t) e^{-(\sigma + j\omega)t} dt$$

$$X(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$s \rightarrow$ complex freq.



→ Provided that $f(t)$ is converging, if $\sigma=0$, Laplace Transform = Fourier Transform
provided, $\sigma=0$ is in region of convergence

Initially, $f(t)$ is non-converging functⁿ
We multiply by decaying functⁿ $e^{-\sigma t}$, so that we convert it into converging functⁿ for some value of σ , then we can apply Fourier Transform

Q. Find Laplace Transform of e^{-at}

Let $f(t) = e^{-at}$

$$X(s) = \int_0^{\infty} e^{-at} e^{-st} dt$$

$$X(s) = \int_0^{\infty} e^{-(s+a)t} dt$$

$$X(s) = \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_0^{\infty} = -\frac{1}{s+a} \left[e^{-(s+a)t} \right]_0^{\infty}$$

$$X(s) = \frac{1}{s+a}$$

$$L.T. [e^{-t}] = \frac{1}{s+1}$$

$$L.T. [e^t] = \frac{1}{s-1}$$

$$1) L.T. (t^n) = \frac{n!}{s^{n+1}}$$

$$2) L.T. (e^{at}) = \frac{1}{s-a}$$

Properties of Laplace Transform

→ Linear

→ Time Shifting

$$f(t) \xrightarrow{L.T.} F(s)$$

$$f(t-t_0) \xrightarrow{L.T.} \int_0^{\infty} f(t-t_0) e^{-st} dt$$

$$e^{-st_0} \int_0^{\infty} f(t-t_0) e^{-s(t-t_0)} dt$$

$$f(t-t_0) \xrightarrow{L.T.} e^{-st_0} F(s)$$

$$\rightarrow f(t) \xrightarrow{L.T.} F(s)$$

$$\frac{d}{dt} f(t) \xrightarrow{L.T.} sF(s) - f(0)$$

$$F(s) = \int_0^{\infty} f(t) \cdot e^{-st} dt$$

$$F(s) = \left[\frac{f(t) e^{-st}}{-s} \right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} f'(t) e^{-st} dt$$

$$F(s) = \frac{f(0)}{s} + \frac{1}{s} L.T. [f'(t)]$$

$$\boxed{\text{L.T. } [f'(t)] = s F(s) - f(0)}$$